

IMPACT ANALYSIS

CR-001: Client-Side Search and Filtering of the Admin Student List

1. Change Summary

CR-001 introduces client-side search and filtering to the administrator's student list.

Administrators will be able to search by registration number, first name, last name, full name, and email. Filtering will be performed on the already-loaded student records.

2. Impact Assessment

Area	Impact	Assessment
Frontend	Low - Moderate	Dashboard search/filtering logic and UI will be added.
Backend	None	Existing API remains unchanged.
API	None	Existing <code>GET /api/students/</code> remains in use.
Database	None	No schema or data changes required.
Authentication	None	Existing authentication remains unchanged.
Authorization	None	Existing administrator protection remains unchanged.
Performance	Low	Filtering is performed locally on the loaded dataset.
CRUD	Low	Edit/Delete must continue using correct <code>student_id</code> .
Deployment	Low	No database migration or new backend endpoint required.

3. Frontend Impact

The main affected area is the administrator dashboard.

Potential changes include:

- Search input.
- Client-side filtering logic.
- Result count.
- Clear Search functionality.
- No-result message.
- Responsive styling.

The implementation must preserve the relationship between each displayed student and its original `student_id`.

4. Backend, API and Database Impact

No backend changes are expected.

The existing:

GET /api/students/

endpoint will continue to provide the complete student list.

No new API endpoint or server-side search is required.

No database schema or data changes are required.

5. Security Impact

The existing authentication and authorization mechanisms remain unchanged.

The student list must continue to be accessible only to authorized administrators.

The search feature must:

- Not bypass existing authorization.
- Not expose additional sensitive information.
- Use normal React rendering.
- Preserve existing backend security controls.

6. CRUD and Functional Impact

Existing administrator CRUD operations must continue to work after filtering.

Particular attention is required for:

- Editing a student from filtered results.
- Deleting a student from filtered results.
- Maintaining the correct `student_id`.

Filtering must only change which records are displayed; it must not change student identity.

7. Performance Impact

The expected performance impact is low because filtering occurs against records already loaded in the browser.

No API request should be generated for every search keystroke.

This approach is suitable for the current expected dataset size. Large datasets may require server-side searching or pagination in a future change.

8. Testing Impact

Manual testing should cover:

- Registration number search.
- First/last/full name search.

- Email search.
- Partial matching.
- Case-insensitive matching.
- Leading/trailing/repeated spaces.
- Result count.
- Clear Search.
- No-result state.
- Edit from filtered results.
- Delete from filtered results.
- Existing CRUD functionality.
- Administrator authentication and authorization.

9. Risk Analysis

Risk	Impact	Mitigation
Wrong student affected after filtering	High	Preserve and use the original <code>student_id</code> .
Search fails because of case/spacing	Medium	Normalize search input and values.
Existing CRUD functionality breaks	High	Perform regression testing.
Client-side filtering becomes slow for large datasets	Medium	Limit current scope; consider server-side search later.
Responsive UI problems	Low	Test supported screen sizes.

10. Overall Impact and Recommendation

Overall Impact: Low

CR-001 is primarily a frontend usability enhancement. It does not require database, backend API, authentication, or authorization changes.

The change is technically feasible and recommended for implementation, subject to formal approval and subsequent testing.

Key implementation consideration: Existing Edit and Delete operations must continue to affect the correct student after filtering.